

Engineering Mathematics III

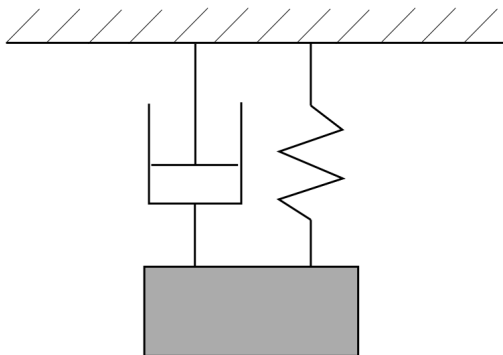
ENGE702

Lecture 8: Second Order Homogeneous ODEs

Second order ODEs

Second order linear homogeneous ODEs with constant coefficients have a number of fundamental physical models. We will look at one called a **Mass - Spring - Damper System**.

This consists of a mass that is attached to a fixed structure via a spring and a damper, as in the following schematic:



Second order ODEs

We need to set up a model to describe the position of the mass at any time.

We will take the equilibrium position of the mass to be $y = 0$, and downwards to be positive displacement.

We will look at the forces acting on the mass.

Hooke's law says that the force from a spring is proportional to its displacement from its equilibrium position, and acts in the opposite direction to the displacement. Thus the force from the spring can be given by

$$F_s = -ky$$

The force from a damper is proportional to the velocity of the motion (again in the opposite direction to the velocity). Thus the force from the damper can be given by

$$F_d = -cy'$$

Second order ODEs

Finally, Newton's law says that the sum of the forces acting on the mass is equal to the mass times acceleration, so

$$F = my''$$

Here F is the sum of all forces, so $F = F_s + F_d$, or

$$my'' = -cy' - ky,$$

rearranging we get

$$my'' + cy' + ky = 0.$$

This is a second order linear homogeneous ODE with constant coefficients, and we will look at the solutions for various values of m , c , and k .

Note in our model that $m, k > 0$ and $c \geq 0$.

Second order ODEs

Case 1: no damping.

In this case our differential equation is

$$my'' + ky = 0$$

Since $m, k > 0$, we can put this in standard form and solve the characteristic equation $\lambda^2 + \frac{k}{m} = 0$

This has solutions

$$\lambda = \pm \sqrt{\frac{k}{m}} i$$

so the system has angular frequency

$$\omega = \sqrt{\frac{k}{m}}$$

Since $-\frac{a}{2} = 0$, the general solution will be

$$y = A \cos(\omega t) + B \sin(\omega t)$$

Second order ODEs

Example: Suppose we have a mass - spring system with mass of 4 and spring constant of 16. The displacement at $t = 0$ is 1 and the velocity at $t = 0$ is 2. Find a function that describes the position as a function of time.

Second order ODEs

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First we know that $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{16}{4}} = 2$. So our general solution is

$$y = A \cos(2t) + B \sin(2t).$$

Differentiating this gives

$$y' = -2A \sin(2t) + 2B \cos(2t).$$

Our initial conditions give $y(0) = 1$ and $y'(0) = 2$.

So we have

$$1 = A \cos(0) + B \sin(0)$$

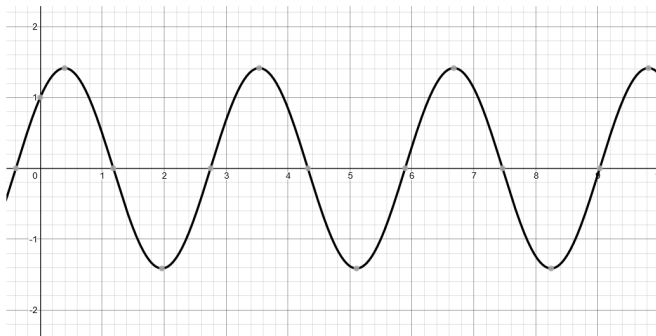
$$2 = -2A \sin(0) + 2B \cos(0)$$

Second order ODEs

Solving this gives $A = 1$ and $B = 1$, so our solution is

$$y = \cos(2t) + \sin(2t).$$

Plotting this gives:



Changing initial conditions will alter the amplitude and phase, but the frequency ω will always stay the same.

Second order ODEs

Now lets see what happens when we add a damper to the system. Our differential equation is now:

$$my'' + cy' + ky = 0.$$

As we saw earlier, the solution type will vary depending on the roots of the characteristic equation.

The roots of the characteristic equation come from the quadratic formula, and are give by

$$\lambda_1 = -\alpha + \beta \quad \text{and} \quad \lambda_2 = -\alpha - \beta$$

where

$$\alpha = \frac{c}{2m} \quad \text{and} \quad \beta = \frac{1}{2m} \sqrt{c^2 - 4mk}$$

Second order ODEs

There are three cases:

Case 1: Overdamping. In this case $c^2 > 4mk$, and there are two distinct real roots.

Case 2: Critical damping. In this case $c^2 = 4mk$, and there is one real root.

Case 3: Underdamping. In this case $c^2 < 4mk$, and there are complex conjugate roots.

We will look at the general solutions of the three cases and give some particular solutions.

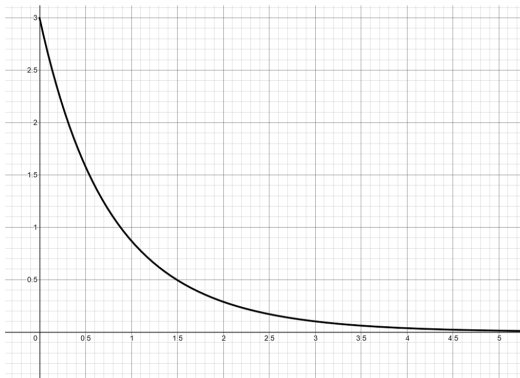
Second order ODEs

Case 1: Overdamping. Here the general solution is

$$y = c_1 e^{-(\alpha-\beta)t} + c_2 e^{-(\alpha+\beta)t}$$

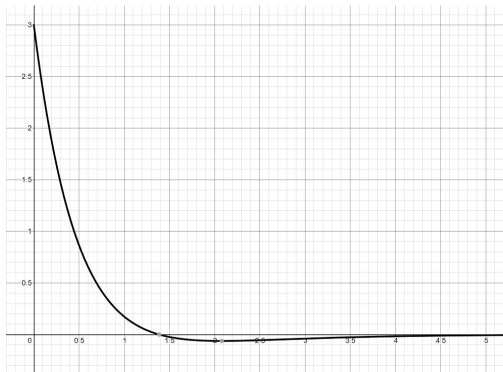
Note that both exponents are negative due to the nature of α and β , so the function will tend to 0 as time increases.

A typical solution looks something like:



Second order ODEs

It is still possible that the mass passes equilibrium in this case, if the initial velocity is large enough, for example:



Second order ODEs

Case 2: Critical damping. Here the general solution is

$$y = (c_1 + c_2 t)e^{-(\alpha t)}.$$

This is the 'borderline' case between oscillatory and non-oscillatory motion. The solution looks much like the overdamped case graphically. It can be thought of as getting to equilibrium 'as fast as possible'.

Second order ODEs

Case 3: Underdamping.

This is the most complicated case. Here our solutions are

$$\lambda_1 = -\alpha + i\omega \quad \text{and} \quad \lambda_2 = -\alpha - i\omega,$$

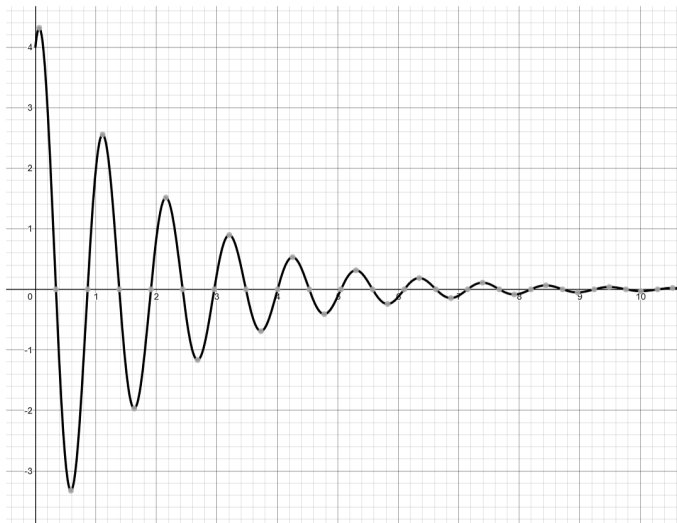
and the general solution is:

$$y = e^{-\alpha t} (A \cos(\omega t) + B \sin(\omega t)).$$

The $\sin$ and $\cos$ terms mean the solution will oscillate about the equilibrium, but the $e^{-\alpha t}$ term means it will tend to the equilibrium as time increases (remember $\alpha = \frac{c}{2m}$ is always positive)

Second order ODEs

The solution will look something like this:



Exercises

Suppose we have a mass - spring - damper system with a mass of $m = 1$, and a damping constant of $c = 4$. Find a value of the spring constant k that will give:

- ▶ underdamped motion
- ▶ critically damped motion
- ▶ overdamped motion

Text Book References

Chapter 2.4